

PRANAVKUMAR RAJYAGURU

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📍 WITTELSBACHERRING 30, 95444 BAYREUTH | 🇩🇪 NATIONALITY: GERMAN



📁 PROFESSIONAL EXPERIENCE

Structural Simulation & Validation Engineer

Oct. 2024 - Present

CYBEX GmbH

- Perform nonlinear finite-element analysis of safety-critical mechanical structures under static, transient, and highly dynamic loading using LS-DYNA.
- Translate regulatory, customer, and system-level requirements into simulation and physical-test load cases, acceptance criteria, and local component loads.
- Develop and validate global and local FE models for structural sizing, stiffness, load-path, and dynamic-response assessment.
- Correlate simulation results with physical measurements from strain gauges, accelerometers, and load cells to improve model fidelity and guide design decisions.
- Investigate structural failure modes, interface loads, and energy-absorption behaviour, combining FE analysis with hand calculations to verify assumptions and identify design improvements.

Master Thesis - Uncertainty Quantification of Safety Systems

Mar. - Dec. 2023

Porsche AG

- Combined experimental data and nonlinear simulations to identify uncertain input parameters in safety-critical systems.
- Applied stochastic modelling, statistical analysis, and Python-based data processing to quantify variation in test and simulation responses.
- Simulated certification-relevant load cases and evaluated the influence of parameter uncertainty on system performance.
- Published the work as “Accounting for and Dealing with Uncertainty Inherent to ATDs in the Design of Occupant Safety Systems” at the Fraunhofer Airbag Conference 2024.

Working Student - Structural Simulation & Dynamic Analysis

Mar. - Dec. 2023

Audi AG

- Developed and validated global and local FE models for structural sizing, stiffness, and dynamic-response assessment.
- Translated full-vehicle load cases into local component and interface loads to support structural and durability evaluations.
- Performed modal and dynamic analyses to investigate vibration behaviour and structural response under test-bench excitation.
- Correlated simulation predictions with measurement data to improve model fidelity and support engineering decisions.

Intern - Structural Simulation

Sep. 2022 - Feb. 2023

Porsche Engineering

- Built and refined finite-element models in ANSA for impact and highly dynamic structural simulations.
- Performed explicit LS-DYNA analyses to investigate structural response, failure mechanisms, and energy absorption under transient loading.
- Evaluated simulation results and communicated technical findings through engineering presentations.

🎓 EDUCATION

M.Eng. Aeronautical Engineering

2020 - 2024

Technische Hochschule Ingolstadt

- Focus: Structural Design, Lightweight Structures, Structural Dynamics, Flight Mechanics, and Aircraft Systems.
- Master's thesis completed in cooperation with Porsche AG.
- Publication presented at the Fraunhofer Airbag Conference 2024.

B.Eng. Mechanical Engineering

2013 - 2017

Gujarat Technological University

🔧 ENGINEERING PROJECTS

Fixed-Wing UAV Hardware Development

2025 - Present

Independent Project

- Develop a fixed-wing UAV through iterative CAD, additive-manufacturing, and hardware-integration cycles, applying experience from assembling, repairing, and flight-testing RC aircraft.
- Modified wing geometry with winglets and reinforced internal structures to improve strength and support the intended aircraft configuration.
- Designed the fuselage and internal layout around the flight controller, battery, ESC, wiring, and payload while managing centre-of-gravity, accessibility, and electromagnetic-interference considerations.
- Introduced removable access features for electronics and servos and simplified component interfaces to improve assembly, maintenance, and design iteration.
- Integrated ArduPilot-compatible hardware, including IMU, GPS, airspeed sensing, telemetry, servos, ESC, and power distribution.
- Used flight testing of RC platforms to investigate oscillatory and overly sensitive behaviour and inform subsequent control-system and configuration iterations.
- Applied topology optimisation and 3D-printing parameter studies to develop landing-gear and structural concepts while addressing support strategy, print quality, and material shrinkage.

Aircraft Structural and Systems Design

Feb. - Aug. 2021

Master's Project

- Developed an integrated aircraft model incorporating aerodynamic, propulsion, structural, and environmental parameters.
- Performed preliminary structural design of the wing, empennage, and landing gear, including mass, centre-of-gravity, thrust, and mission-performance assessments.
- Analysed aircraft dynamic behaviour and developed flight-control and autopilot models in MATLAB/Simulink.

Formula Student - Structural Design & Aerodynamics

- Developed aerodynamic component concepts using CAD and evaluated their aerodynamic and structural behaviour through CFD and coupled structural analysis.
- Performed stiffness assessment and topology optimisation of aerodynamic components while coordinating design decisions with the manufacturing team.

Swissloop Tunneling - Structural Design

ETH Zürich Student Team

- Contributed to structural design and CAE activities for the team's tunnelling system within a multidisciplinary engineering environment.
- Supported the development of the team's award-winning design for the Not-a-Boring Competition.

CERTIFICATES

- EU Drone Licence A1/A2/A3, LBA.
- Aerospace Structures and Materials, TU Delft.
- Applying Systems Engineering to the Design Process, University of Colorado Boulder.
- Introduction to Model-Based Systems Engineering, Siemens.

SKILLS

- **Structural Analysis:** Nonlinear FEA, structural dynamics, modal analysis, impact simulation, load-path analysis, model validation, and simulation-test correlation.
- **CAE:** LS-DYNA, ANSA, Ansys, NASTRAN, ABAQUS, HyperMesh, and OptiStruct.
- **CAD and Prototyping:** CATIA V5/V6, Siemens NX, topology optimisation, and additive manufacturing.
- **UAV Systems:** ArduPilot, flight-controller and sensor integration, GPS, telemetry, airspeed sensing, ESCs, servos, and power distribution.
- **Programming and Analysis:** Python, MATLAB/Simulink, shell scripting, and data analysis.

LANGUAGES & RELEVANT ACTIVITIES

Languages German: C1, English: B2

Activities Wildlife-rescue drone pilot; model-aircraft and gliding enthusiast